ImU: Physical Impersonating Attack for Face Recognition System with Natural Style Changes

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***Abstract*—This paper presents a novel physical impersonating attack against face recognition systems. It aims at generating consistent style changes across multiple pictures of the attacker under different conditions and poses. Additionally, the style changes are required to be physically realizable by make-up and can induce the intended misclassification. To achieve the goal, we develop novel techniques to embed multiple pictures of the same physical person to vectors in the StyleGAN's latent space, such that the embedded latent vectors have some implicit correlations to make the search for consistent style changes feasible. Our digital and physical evaluation results show our approach can allow an outsider attacker to successfully impersonate the insiders with consistent and natural changes.**

I. Introduction

Deep learning models have been widely used in face recognition systems due to their impressive performance [1], [2], [3], [4]. For example, with sufficient training data, sophisticated model architectures, and advanced training strategies, existing face recognition models can achieve more than 99% accuracy. With the emerging Machine Learning as a Service (MLaaS) provided by large vendors [5], [6], [7], [8], it also becomes easier for non-experts to deploy their own face recognition systems. Despite their impressive performance, recent research has revealed that face recognition models are vulnerable to adversarial attacks that can mislead an input image to a pre-defined target. Most of the existing attacks reside in the cyber space [9], [10], by directly adding *invisible* (stealthy) perturbations to the digital images, and feeding them into the classification models. However, most of these digital attacks rarely threaten the systems in the real world, due to the fact that the perturbation noises cannot be physically implemented or captured by the camera.

To extend the attacks to the physical world, a few impersonating attacks utilize more *visible* noises yet confined in small local (masked) areas (*e.g.*, eyeglass frame [11] or stickers [12]). For example, AdvGlass [11] uses the traditional digital attack to generate unbounded adversarial noises in the eyeglass frame area. The attacker then prints out the patch and attaches it to a pair of glasses. Wearing this pair of glasses can make the model recognize the attacker to the target person.

However, adversarial noises of existing physical impersonating attacks (*e.g.*, noticeable adversarial accessories) are still too distinguishable from natural faces to be stealthy.[1] Also, accessories are not allowed in many security-related situations such as the photos for visa application [13]. Another limitation of existing physical impersonating attacks is that they lack consistent effectiveness across faces captured at different poses, as they usually attack only one face or faces with almost the same poses.[2]

Different from the definition of stealthiness in existing attacks, we argue that the overall perturbation naturalness (such as whole-face style changes consistent across different poses) instead of bounded perturbation at the pixel level can provide better stealthiness. For example, at a security check location, a person wearing a pair of weird glasses is more ostentatious and suspicious than a person wearing daily makeup. To this end, we utilize StyleGAN, which has been widely used in image editing tasks [14], [15] due to the embeddability (i.e., the ability to invert a given image to a latent value of the StyleGAN) and semantic editability (i.e., style editing) of its latent space. Although existing methods [15], [16] can highly effectively mutate the style of a given image, they can hardly achieve consistent style changes across different images from the same physical person (potentially with different poses), which is key to achieving an effective physical impersonating attack.

In this paper, we propose a novel impersonating attack based on StyleGAN. Given a set of pictures of the attacker, under different conditions and having different poses, our attack aims to produce consistent style changes that are physically realizable and can flip the classification result to a target person. In order to achieve the goal, we need to embed the set of images, namely, generating the latent vectors for these images. The StyleGAN does not have any concept of face identity and hence the embedded latent vectors may not have any strong correlations although they belong to the same physical person. As a result, finding a consistent style change for this set of images (from the same physical person) is no different from finding it for a set of arbitrary images, which is difficult. We have two key observations that allow us to

[1]We conducted a user study via Amazon MTurk with 240 users. About 70% more users consider AdvGlass's images are more noticeable than ours.

[2]Although there exist some spatial transformations to ease this problem for traffic signs, how to apply them for face images (e.g., from the frontal face to the side face) is still largely open.

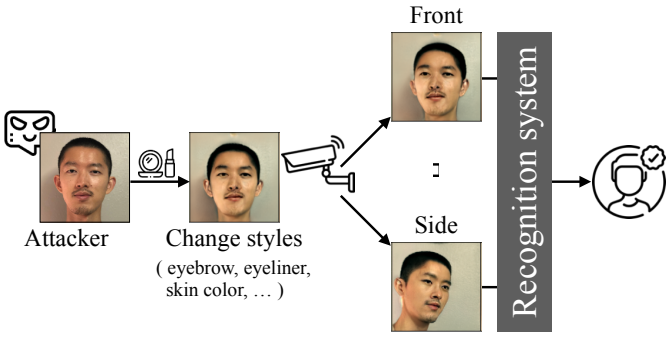

Fig. 1: Attack scenario.

address the problem. First, we find that poses of generated faces by the StyleGAN can be controlled by injecting specific noises at certain style blocks. Second, given an attacker's image with pose p, if we can find the corresponding noise and then enforce such during embedding, we are able to force the StyleGAN to establish some implicit connections for the multiple images of the attacker, like recognizing that they belong to the same person. As such, we can find a consistent style change for these *properly embedded* images (i.e., images generated by the embedded latent vectors) that enables the impersonating attack.

In summary, we have the following contributions:

- We propose a novel impersonating attack approach based on StyleGAN. It supports both white-box and black-box settings.
- We study the effects of random noises injected in StyleGAN. Based on our new findings, we design a novel embedding approach to faithfully project real attacker's images with different poses into the latent space.
- We develop a batch-based attack method with statistical constraints to generate consistent, physically applicable and stealthy style changes for images of the same person with different poses.
- We build a tool ImU (Physical Impersonating Attack for Face Recognition System with NatUral Style Changes) and evaluate it on 10 large models pre-trained on 4 large-scale face recognition datasets and 2 commercial services. Digital evaluation results in both white-box and black-box settings show our approach can generate consistent natural style changes for faces with different poses. In the white-box setting, our approach can achieve the highest attack success rate even with the physical-world simulation (*i.e.*, adding noises to and cropping and rescaling the images). In the black-box setting, our approach can achieve 16x higher attack success rate compared to the existing methods. After physically applying the generated changes, attackers can successfully conduct the physical impersonating and fool the classifiers. We also show attackers can physically attack the online commercial services. On models trained with different adversarial training strategies (even with the adaptive defense), our approach still yields high attack success rate.

II. Physical Impersonation Attacks

In this section, we first introduce our threat model and then summarize four requirements of realizing physical impersonation attacks. Afterwards, we show existing methods fail in fulfilling them and showcase our results.

A. Threat Model

Attack Scenario. Figure 1 shows our attack scenario. Our subject model is a face recognition system based on a neural network classifier or verifier.

A classifier is trained on face images from a set of N authorized people. Given one image, the system predicts the label from N. When recognizing a person, the system requires taking multiple photos with different poses. The attackers are *outsiders* not among the N people (*i.e.*, out-of-distribution) who want to impersonate some target people (*e.g.*, label 1) among the N people (*i.e.*, in-distribution).

A verifier is trained to extract features of given images so that images of the same person have smaller feature distances (*e.g.*, higher cosine similarity score) than those of different people. During recognition, if the similarity score of two images (*i.e.*, the attacker's and target person's images) is larger than a pre-defined threshold, they are considered as one person.

Specifically, the outsider attacker aims to *physically* impersonate the target people by changing the styles. With the changed styles, the attacker's images captured with different poses should be recognized as the target person by the system. More importantly, the style change should not be ostentatious or suspicious.

Attackers' Capability. We assume that attackers can utilize public face data (*e.g.*, CelebA [17]) to train generative models (*e.g.*, StyleGAN [18]) . The training data can be non-overlapping with the subject model's training data. For face classifiers, attackers need no access to the target person's image. For face verifiers, attackers have the target person's image. In the white-box setting, attackers can access the internals of the subject model and thus can use a gradient-based method to conduct the attack. In black-box attack on a classifier, attackers treat the model as a black box and use a query-based method. In black-box attack on a verifier, attackers conduct a transferable white-box attack against surrogate verifiers. Our attacker's capability is consistent with existing literature [11], [19], [20], [21], [22]. Attackers should be able to apply makeup by themselves or ask some professionals to do that.

B. Requirements of Physical Attacks

We follow the standard way to conduct physical attacks: the attacker first generates the adversarial image in cyberspace, and then tries to physically realize it in the real world. In the following, we summarize four challenging requirements for conducting physical attacks.

Requirement 1: ability of using outsider attacker's images. A physical attack uses a real, outsider attacker, and an attack

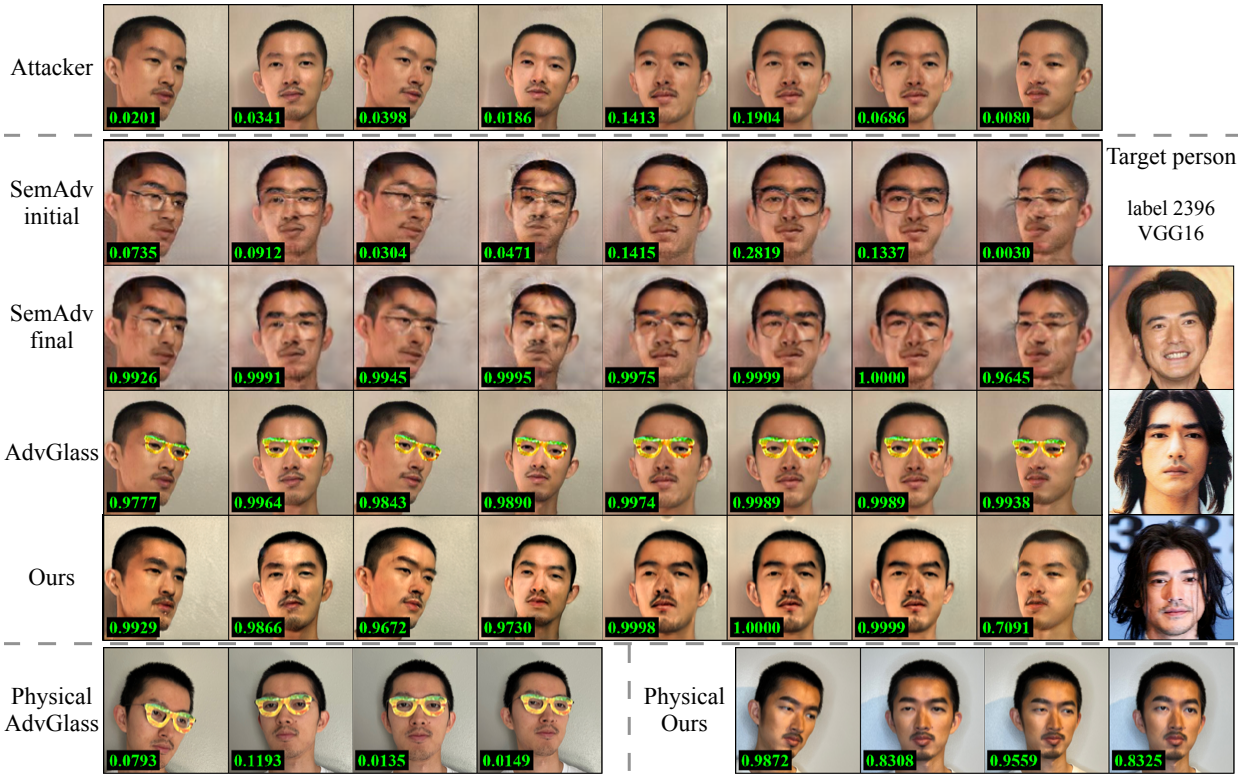

Fig. 2: Digital/physical examples of different attacks. The target person is label 2396 of VGGFace dataset and the subject model is VGG16. Three ground truth images of the target person are displayed in the rightmost column. The first row shows the images of an attacker with different poses. The number in each image denotes the confidence. AdvGlass's results are more obvious/ostentatious and less effective than ours. SemAdv was proposed as a digital attack. Appendix has another example where the same attacker impersonates an actress.

method should be able to use real people's images. For example, based on one real image of the attacker, the attack method should return an updated/generated image which can 1) fool the subject model, and 2) guide the attacker to physically apply the changes.

Requirement 2: physical applicability of the adversarial changes. We need to be able to physically apply the adversarial changes for a physical attack. For example, we cannot exactly map each pixel in the digital image (*e.g.*, one pure red pixel) to a certain dot of the face. In addition, sharp change of colors usually cannot be captured by the camera.

Requirement 3: stealthiness of the adversarial changes. In the physical world, we also aim to keep the stealthiness of the attack, preventing the adversarial changes from being too suspicious in humans' perspective. For example, it would be too ostentatious to have a half-red and half-green face.

Requirement 4: effectiveness across different poses and views with the same adversarial change (*i.e.*, consistency). Finally, it is necessary that the attacker with the style changes should be able to consistently intrude the subject classifier at various poses. The reason is that when captured by the camera, the attacker cannot exactly ensure a certain pose.

C. Limitations of Existing Methods

AdvGlass [11]. AdvGlass is a representative physical attack in literature. It uses traditional unbounded digital (pixel-based) attacks (*e.g.* PGD [9]) and confines the optimization in certain areas (*e.g.*, eyeglasses). Specifically, given an image x of the attacker, AdvGlass uses an eyeglass frame mask M to constrain the optimization of adversarial perturbations such that $x \odot (1 - M) + \odot M$ is recognized as the target person. M only contains 1 (for the eyeglass frame area) and 0 elsewhere. It also enhances the physical applicability by encouraging the colors used by the adversarial perturbations to be close to a set of printable colors. Then, the attacker can print out $\odot M$ and attach it on a real eyeglass frame. To robustify the adversarial glasses, AdvGlass uses a set of images with *slightly* different poses (almost frontal views) and thus the glass frame mask position is almost fixed without any perspective transformations.[3]

There are three drawbacks of AdvGlass. First, accessories like eyeglasses are not allowed in many security-related circumstances such as photos used in visa applications [13]. Second, the weird adversarial glasses are still too ostentatious

[3]In practice, we extend AdvGlass by automatically resizing and rotating the mask and noises to fit the different poses.

and suspicious to be stealthy. From the fourth row of Figure 2, we can observe that the eyeglasses indeed have strange colors and are not natural. Third, although AdvGlass uses a set of images with *slightly* different poses, the glass frame mask position is still almost fixed frontal views without any perspective transformations. Consequently, with printed eyeglasses, the attacker may not be able to successfully launch the attack with different poses. The results are shown in the left part of the last row in Figure 2. None of them are successfully recognized as the target person[4].

SemAdv [19]. Applying usual digital (pixel-based) attacks in the physical case is extremely challenging, and SemAdv is a potential choice as it is able to edit images on the attribute level. Specifically, SemAdv first uses a single-attribute editor network (*i.e.*, a pair of encoder E and decoder D) to change one feature c of the input image x (*e.g.*, adding/removing glasses). For example, x is a face without glasses (*i.e.*, the original feature vector $c_0 = 0$), $D(E(x; 1-c_0))$ will add a pair of glasses to x. To make the edited image recognized as the target person, SemAdv further optimizes a coefficient tensor to interpolate between the internal encoding of the attacker's raw image and the edited image.

However, applying SemAdv physically still has several limitations. First, as shown by the third row of Figure 2, the style changes at different poses are not consistent. This is because SemAdv can only attack one image (frontal view). Second, SemAdv is not physically applicable as the adversarial noises are dispersed across the image instead of concentrating on the faces (see Figure 29 in the appendix). Third, SemAdv's capability of using real images is closely dependent on the capability of D and E (*i.e.*, StarGAN [23] in their paper). As such, SemAdv can only work for in-distribution data (*i.e.*, frontal view faces very similar to training data) because StarGAN is less generalizable on unseen data than some more sophisticated GANs (*e.g.*, StyleGAN).

AdvMakeup [20]. Similar to the eyeglasses mask in AdvGlass, AdvMakeup also uses a mask M to only modify the orbital area. AdvMakeup attacks verifiers. For *one* target image x_t, it trains a model G to generate changes $= G(x_a)$ for *one* image x_a of an attacker such that $x_a \odot (1 - M) + \odot M$ and x_t have a small feature distance. Figure 3 shows the results of ours and AdvMakeup. Because of the mask-based modification, the patched area has visual disparities (thus less stealthy and natural). Also, AdvMakeup attacks each image separately, and there are hence no explicit constraints on the consistency.

D. Our Digital and Physical Examples

Table I summarizes to what extent different attacks satisfy the four requirements. We aim to simultaneously satisfy the four requirements, and some examples are shown in Figure 2. Compared to existing methods, our results (fifth row) are consistent among different poses, natural-looking, and thus stealthier. Different from existing physical attacks, we claim that the small size of adversarial noises is not the necessary condition of a stealthy attack, and instead the stealthiness in the physical world is highly related to the naturalness of the adversarial changes. One intuitive example is changing an attacker's facial style with cosmetics. In order to show the effect of our physical attack, we asked a cosmetology student to apply makeup on the attacker according to the generated adversarial images. The results are shown in the right part of the last row. All of them are recognized as the target person with much higher confidence compared to AdvGlass. To check if the confidences of our physical examples are high enough to break the face classifier, we collected a set of images of the target person online (not in the training dataset) and tested their confidences. All the downloaded images are correctly classified as the target person. If we set a confidence threshold for these images, the threshold should be smaller than 0.5853 so as to ensure 60% accuracy. In this case, all of our physical examples can pass this threshold.

TABLE I: Summary of different attacks

Challenges	AdvGlass	SemAdv	AdvMakeup	Ours
1) Real attackers	Yes	No	Yes	Yes
2) Physical applicability	Partial	No	Yes	Yes
3) Stealthiness	Partial	-	Partial	Yes
4) Consistency	Extended by us	No	Partial	Yes

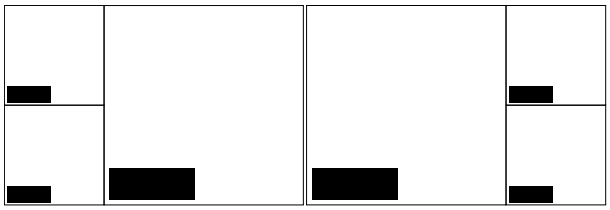

Fig. 3: Examples of ours (left) and AdvMakeup (right). The subject model is a verifier FaceNet pre-trained on VGGFace2. The target image is the third target image in Figure 2. Please zoom in for a better inspection.

III. Background: StyleGAN

In this section, we introduce StyleGAN, which is an important component of our approach. Traditional GANs were proposed to learn a mapping from a latent space $\mathcal{Z}$ (*e.g.*, Gaussian distribution) to another space $\mathcal{X}$ (*e.g.*, Human face distribution). However, the generated image's quality and resolution still have a lot of room to improve. In order to generate better images, StyleGAN was proposed by [18]. Unlike traditional GANs with one latent space $\mathcal{Z}$, StyleGAN utilizes another intermediate latent space $\mathcal{W}$ mapped from $\mathcal{Z}$. This intermediate latent space will be further transformed into the styles allowing gradual adjustment of styles at different granularities to forge the final image.

Figure 4 shows the simplified architecture of StyleGAN which can be partitioned into two parts: a mapping network F (left) and a synthesis network G (middle). The original

[4]AdvGlass was also reported to have a poor physical attack performance on complicated models [20]. The models physically attacked in the AdvGlass paper recognized only $\leq$ 143 identities [11], whereas VGG16 here recognizes $\geq$ 2.6K identities.